THE CLASSIFYING STACK [B/G] IS A GERBE OVER B

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-groupoids.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original lemma 044R: sources/stacks-chow/tags.txt, source lines 5410–5410. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 06PE: sources/stacks-chow/tags.txt, source lines 8568–8568. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let B be an algebraic space over S. Let G be a group algebraic space over B. Endow B with the trivial action of G. The morphism*

$$[B/G] \longrightarrow \mathcal{S}_B$$

(Lemma 044R) turns $[B/G]$ into a gerbe over B.

Proof. Immediate from Lemma 06PE as the morphisms $B \to B$ and $B \times_B G \to B$ are surjective as morphisms of sheaves. □

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